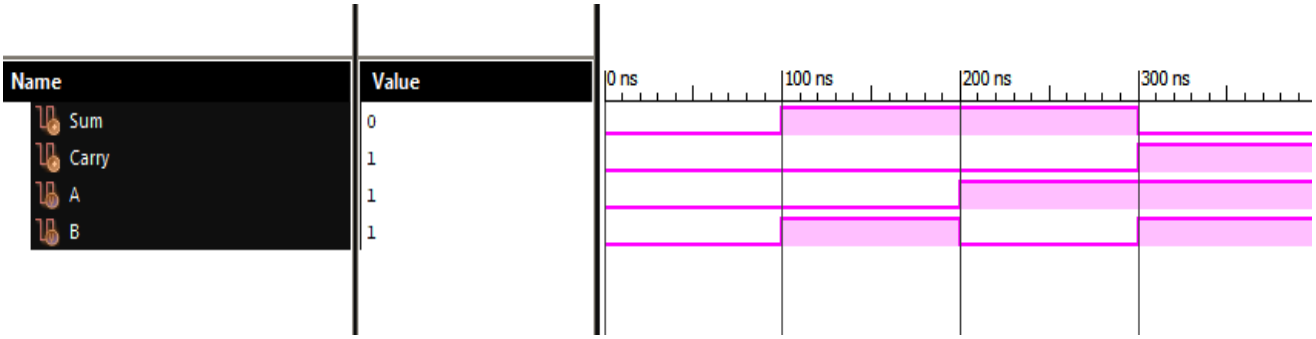
VERILOG PROGRAM FOR HALF ADDER (DATA FLOW MODELING)

```
module half_adder (A,B,Sum,Carry);
    input A, B;
    output Sum, Carry;
    assign Sum = A ^ B;
    assign Carry = A & B;
endmodule
```

OUTPUT FOR HALF ADDER



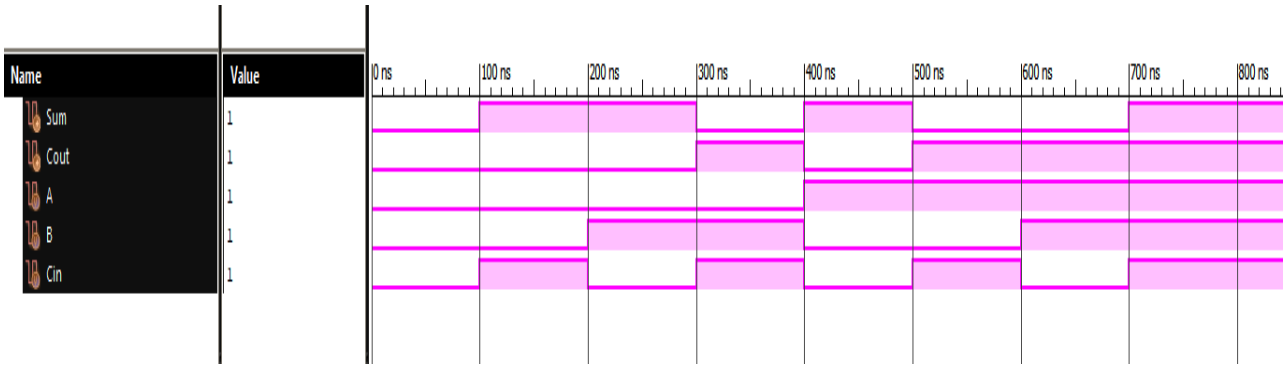
VERILOG TEST FIXTURE

```
half_adder uut (
                .A(A),
                .B(B),
                .Sum(Sum),
                .Carry(Carry)
        );
        initial begin
                A = 0;
                B = 0;
                #100;
                A = 0;
                B = 1;
                #100;
                A = 1;
                B = 0;
                #100;
                A = 1;
                B = 1;
                #100;
                end
```

VERILOG PROGRAM FOR FULL ADDER(DATA FLOW MODELING)

```
module full_adder (
    input A,
    input B,
    input Cin,
    output Sum,
    output Cout
);
     assign Sum = A ^ B ^ Cin;
    assign Cout = (A & B) | (B & Cin) | (A & Cin);
endmodule
```

OUTPUT FOR FULL ADDER



VERILOG TEST FIXTURE

```
full_adder uut (
                .A(A),
                .B(B),
                .Cin(Cin),
                .Sum(Sum),
                .Cout(Cout)
        );

        initial begin
                // Initialize Inputs
                A = 0;
                B = 0;
                Cin = 0;
                #100;
```

```
        A = 0;
        B = 0;
        Cin = 1;
        #100;

        A = 0;
        B = 1;
        Cin = 0;
        #100;

        A = 0;
        B = 1;
        Cin = 1;
        #100;

        A = 1;
        B = 0;
        Cin = 0;
        #100;

        A = 1;
        B = 0;
        Cin = 1;
        #100;

        A = 1;
        B = 1;
        Cin = 0;
        #100;

        A = 1;
        B = 1;
        Cin =1;
        #100;
    end

endmodule
```